ResearchGate

See discussions, stats, and author profiles for this publication at: https://www.researchgate.net/publication/348192061

Human-in-the-loop enhanced COVID-19 detection in Transfer Learning based CNN Models

Chapter · January 2021

CITATIONS
0

READS
52

2 authors, including:

Ferhat Ozgur Catak
Simula Research Laboratory

41 PUBLICATIONS **145** CITATIONS

SEE PROFILE

Some of the authors of this publication are also working on these related projects:

Project Machine Learning Based Cyber Security View project

Project Privacy Preserving Machine Learning View project

All content following this page was uploaded by Ferhat Ozgur Catak on 04 January 2021.

The user has requested enhancement of the downloaded file.

Human-in-the-loop enhanced COVID-19 detection in Transfer Learning based CNN Models

Ferhat Ozgur Catak and Kevser Şahinbaş

Abstract Recently, many countries have concentrated on protecting human health and struggle with COVID-19 outbreak. Now that the virus is showing up in many places worldwide and the numbers of infected people are increasing rapidly, the governments of some countries are going to have to act quickly. The diagnostic tools are expensive, and the diagnostic kits can be challenging to obtain in many countries which makes it more difficult to stop the epidemic. Therefore, we need a fast and straightforward diagnostic tool based on AI technology that will save time, money and resources in the prevention and control of the virus. To achieve this problem, researchers are developing X-Ray scan images based COVID-19 detection models which are based on convolutional neural networks and transfer learning algorithms. These models are trained on visual pictures in order to predict the presence of the virus in the sample. On the other hand, predictive modelling can make some false predictions like false-negative which means that the sample may not be infected with COVID- 19, but it will show positive COVID-19. It is important to remark that the wrong diagnostic will lead to an incorrect approach to the treatment and prevention of the virus. For example, if a patient is infected with COVID-19, but the test is negative for the virus, the patient can lose their life. Such an incorrect diagnosis will be very costly for the government. For this reason, model performance and the reliability of the classification result should be considered together. Thus we recommend applying to a medical doctor for low quality estimates by using uncertainty.In this study, we propose uncertainty quantification enhanced transfer learning-based convolutional neural network models for predicting the presence of COVID-19 in X-Ray scan images publicly available on a GitHub repository. During the training phase, we used 278 positive and 66 negative COVID 19 X-rays and 39 positive and 30 negative images during the testing phase. We used VGG16, VGG19, ResNet, DenseNet, and Inception V3 pre-trained deep CNN models to cope with

Ferhat Ozgur Catak
Simula Research Lab. Oslo, Norway, e-mail: ozgur@simula.no

Kevser Şahinbaş
Istanbul Medipol University, Istanbul, Turkey e-mail: ksahinbas@emedipol.edu.tr

the complex nature of the images that require deep architecture. Findings from the results demonstrate that the pre-trained VGG16 model can detect COVID-19 from non-COVID-19 cases with the highest classification performance among the other proposed models with 75% accuracy and can be used as an additional tool in the radiology department.

Key words: COVID-19, transfer learning, CNN, X-Ray, Uncertainty Quantification

1 Introduction

COVID-19 is a respiratory virus in the human (pulmonary) lung with the highest infectious potential of all respiratory viruses. It can be transmitted by aerosol particles and by respiratory droplets (e.g., from coughing, sneezing from an infectious agent). The transmission rate of COVID-19 is very high, and the disease can be fatal if it is not treated in time. The clinical studies based on COVID-19 diagnosis can be time-consuming and expensive, and it is challenging to detect in the population (as in many other viral diseases). This issue makes it extremely important to identify the disease with cheap and fast diagnostic tools that can be used in hospitals. Artificial intelligence (AI) is one of the most promising tools for the diagnosis of this disease. AI can detect the presence of COVID-19 from the X-Ray images (as in the case of the present study), and it can also be used to perform the diagnostic steps and to assess the severity of the disease in the population.

There are some limitations in machine learning-based approaches for the diagnosis of COVID-19, such as computational complexity. These limitations can be overcome with the use of the transfer learning models, which allow combining the information obtained from the different data sources to improve the accuracy of the machine learning models. This type of transfer learning has been applied in many fields (e.g., medicine, computer vision, speech recognition) where the data is relatively large (e.g., thousands of images) and the training data is relatively small (e.g., few hundred images). Transfer learning models are also useful for the diagnosis of COVID-19 since they can be applied to a large number of data sources (e.g., X-Ray images), and they are very efficient.

On the other hand, AI-based models for the prediction should be reliable, and we need to quantify the quality of the model's predictions which is essential for making reliable recommendations to the patients. The uncertainty is a promising issue for the reliability of the machine learning models, and the future of AI-based diagnostics is still a matter of many years. For this reason, the present study is the first one to use AI for the diagnosis of this disease with uncertainty quantification. We believe that the present study can serve as a basis for future investigations in this field.

We have summarized some studies on transfer learning-based diagnosis of COVID-19 using chest X-rays. Recently, many radiology images are widely used to detect COVID-19. Wang [1] introduced an artificial intelligence that could diagnose coronavirus. The developers who named artificial intelligence COVID-Net

which obtained 92.4% accuracy. The developers, who collected 5,941 x-ray images of thousands of patients with viral and bacterial infections, uploaded this data to the database of COVID-Net. As a result of this great devotion, COVID-Net only examined lung x-rays and was able to diagnose coronavirus based on these data in the database. Ucar and Korkmaz [2] present SqueezeNet model network design that is called as COVIDDiagnosis-Net a COVID-19 diagnosis with Bayesian optimization additive in X-rays images. In the study, a deep learning architecture-based rapid diagnosis system was proposed using thousands of internationally shared lung X-ray images by focusing on chest radiography imaging and analysis due to abnormalities in chest X-ray images of Covid-19 infected patients. The developed system can only quickly determine the diagnosis of Normal (not infected), Pneumonia (not from Covid-19) or Covid-19 Infected with lung x-rays. Ioannis et al. [3] presents Convolutional Neural Network (CNN) architectures with transfer learning. They obtain a 97.82% overall accuracy in the detection of Covid-19. Hemdan et al. [4] demonstrate deep learning models that named as COVIDX-Net to classify COVID-19 in X-ray images. Narin et al. [5] presented pre-trained ResNet50, InceptionV3, and Inception-ResNetV2 convolutional neural network-based models with 98% accuracy, which is the best of the pre-trained ResNet50 model to detect COVID-19 cases on chest X-ray.. Zhang et al. [6] present a model that predict the prognosis of COVID-19 patients and distinguish patients SARS CoV-2 against other viral infections. Xu et al. [7] proposed ResNet model that is a pre-trained deep CNN ResNet model to detect COVID-19 patients by using CT images thatand has a 86.7% accuracy performance rate.

In our previous work, we showed that the transfer learning-based CNN models can be used to predict the COVID-19 based on X-Ray images, and this work is based on the same experimental data (i.e., the same set of images from which we extracted) [8].

In our best knowledge, the current literature is based on the same approach to detect the COVID-19 based on X-Ray images and CNN models. On the other hand, we believe that a medical doctor should enhance the ability of the model to predict the prognosis of the COVID- 19 in some cases. The current deep neural network models predict the label (positive or negative). To accurately find the COVID-19 patients, the model's predictions have to be reliable. A traditional neural network predicts the label based on softmax probability outputs. Recently, a new paradigm in deep learning has been proposed to calculate the reliability (or uncertainty) of the prediction. The next step would be to calculate the confidence of the COVID-19 model based on the predicted probability. This approach has been applied to a variety of different imaging techniques, but it will be necessary to improve the accuracy of the COVID-19 model by incorporating uncertainty quantification to assess the predictors. The most important feature of our research is that we will be able to improve the accuracy of the COVID-19 model by applying uncertainty quantification to the prediction models.

In this book chapter, we will describe the basic concepts and use of uncertainty quantification to derive reliable predictions for COVID-19. We will show the importance of the uncertainty in machine learning models for the classification per-

formance of the diagnostic model. We will also discuss the practical application of uncertainty quantification for the COVID-19 diagnosis. Section 2 includes Convolutional Neural Network (CNN), transfer learning concept and pre-trained algorithms. In Section 3, model uncertainty in Bayesian deep learning is explained. Data Collection and explanation are given in Section 4. Section 5 contains system model and simulation results. Lastly, we conclude the chapter in Section 7.

2 Convolutional Neural Network (CNN)

There is increasing demand in many areas such as medical image processing, processing of large data sets, processing digital images, and supporting the movement of autonomous cars [9]. CNN studies have been carried out since the late seventies [10].CNN introduced by LeCun in 1989 for computer vision applications, is a type of multi-layer feed forward ANN [11]. CNN is a class of deep and feedforward artificial neural networks that is frequently used in image processing and computer vision problems and is considered the latest technology for image classification [12]. It is designed to mimic the structure of the animal visual cortex [13] .The convolutional neural network (CNN) is a deep learning technique used to represent data using the hierarchical layers of an image (or other input data). This feature makes CNN the right choice for many tasks, such as face detection, object detection. CNN is used in a wide variety of applications and is a powerful tool for classification and image segmentation. The convolution layer represents the input image in a 3 -dimensional space, while the activation layer is responsible for determining the output of the network. The convolutional layer is the central part of the CNN, and the network architecture can be described as follows: a convolutional layer with two input layers, a max-pooling layer, and a fully connected layer. The feature map is created by applying the high and low-level kernels in each of the input.

In Equation 1, the convolution process is shown.

$$S(i,j) = (I \times K)(i,j) = \sum_m \sum_n I(m,n)K(i-m, j-n) \tag{1}$$

Where m and n are the kernel size, in the above equation, the kernel is the input of the pooling layer; the feature map is the output of the pooling layer. Pooling layer is generally applied between convolution layers. The main feature of the pooling layer is to reduce the number of computation steps needed to reduce the amount of the feature map. Another reason for the use of pooling layer is to increase the effectiveness of the network by removing the dominant features from the model. There are different types of pooling layers, and maximum and average pooling layers are the most common in the CNN. In the fully connected layers, the nodes are connected to the previous layer in the network.

Pooling layer discards unnecessary and redundant attributes, which reduces network computing cost and therefore makes the network more efficient [14].

2.1 Transfer Learning

This section provides some of the well-known deep learning image classifiers that are needed to apply the clinical aim of the detection of COVID-19 framework. Transfer learning is the reuse of information learned by a network in another data set to improve the performance of another network [15]. Classical machine learning techniques can work efficiently only when there is sufficient and appropriate training data. In real life, it is difficult to find training data that covers all situations, and in some cases, one of the most significant difficulties encountered by researchers is the limited number of the available dataset [16]. It is difficult to find training data, such as in the analysis of medical data. Besides, with the change of conditions, the training data at hand can become outdated and become unavailable. In such cases, all the approaches that try to reduce the need for training data for machine learning by allowing information exchange between similar tasks are called Transfer Learning that provides the training of data with a small dataset and needs fewer calculation costs [17]. It is the most significant advantage of using transfer learning. In this paper, we structured deep CNN based VGG16, VGG19, InceptionV3 and DenseNet models for the classification of COVID-19 Chest X-ray images to detect COVID-19 classes.

2.1.1 VGG16 and VGG 19

One of the deep learning algorithms is the Visual Geometry Group Network (VGG) architectural structure by Simonyan and Zisserman [18]. It is a model developed by Oxford University to give the best results compared to other algorithms. In this model, the number of convolutions has been increased to 13 for better results. Three full connected structures were used. It is a model consisting of a total of 41 layers with all layers. The size of the images in the input layer is $224 \times 224 \times 3$. As in AlexNet architecture, the classification process is performed in the result layer, and the result is produced [18]. The deep network architecture is classified into two main categories; namely VGG16 and VGG19. These models have different depths and layers [19]. VGG19 is deeper, and the number of parameters is larger and more expensive than VGG16 to train the network.The VGG19 architecture includes approximately 138 million computational parameters [20]. Figure 1 indicates the VGG16 architectures [21].

2.1.2 Inception V3

The input images are given as $224 \times 224 \times 3$ and consist of 22 layers of depth. GoogLeNet consists of modular structures connected in parallel instead of sequentially ordered convolution and pooling layers. GoogLeNet is a deep learning network created by Christian Szegedy et al. in 2015 using the ImageNet dataset. This model, with 48 layers depth takes the image input dimensions as $299 \times 299 \times 3$. Inception

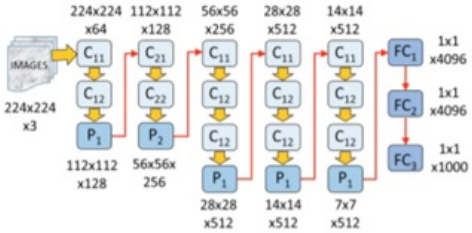


Fig. 1: VGG16 Architecture
[21]

V3 won 2014 Image net challenge with 93.3% top-5 accuracy [22]. Figure 2 demonstrates the compressed architecture of the InceptionV3 convolutional neural network [23].

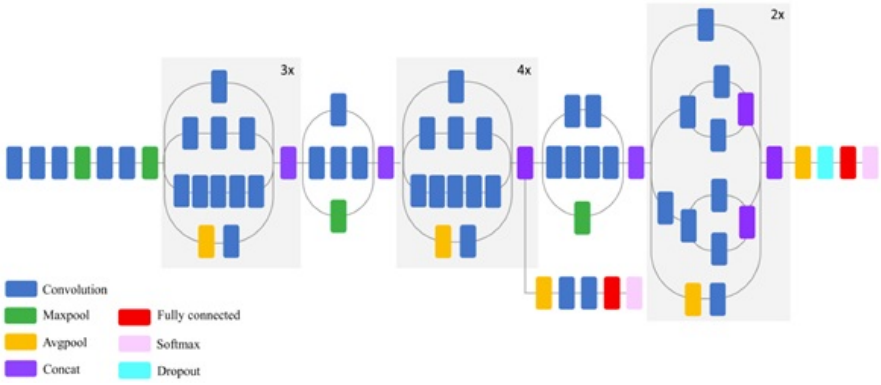


Fig. 2: InceptionV3 architecture
[23]

In this network, fully connected layers are replaced by the average pooling layer. Since fully connected layers contain a large number of parameters, this modification process significantly reduces the number of parameters [24].

2.1.3 DenseNet

This network is designed to overcome the vanishing gradient problem caused by the depth of the network. In particular, the connection architectures of all layers are used to ensure the maximum flow of information between layers . In this configuration,

each layer receives entries from all previous layers and passes its attribute maps to all subsequent layers. Attribute maps are combined on each layer to transmit information to layers after previous layers. This network architecture eliminates the need to learn redundant information and, accordingly, the number of parameters is significantly reduced. It is also effective in protecting information thanks to the linking feature of all layers. In the study by Huang et al., the network performed very well for classifications made with a small training data set. Figure 3 shows a compressed image of DenseNet [25].

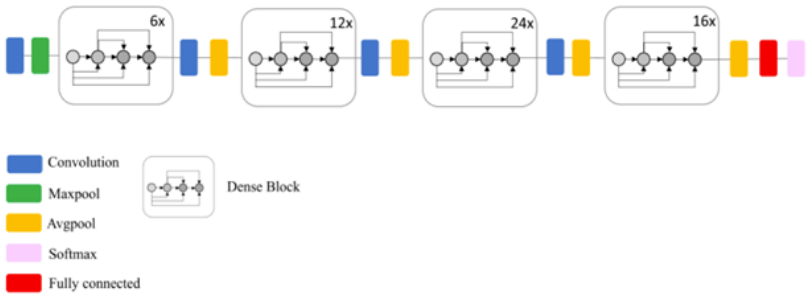


Fig. 3: DenseNet architecture
[25]

2.1.4 Resnet

In 2015, ResNet was the winner of the contest with a 3.57% error in the ImageNet data set in the ILSVRC fight. He et al. [26] instead of teaching the layers the functions that are not distinctive in ResNet, they reformulate the remaining functions as learning by referring to layer inputs. Optimizing residual networks now ensures accuracy at increasing depths. This was revealed in tests with Resnet [23].

Under normal conditions, the part from the input to the output is expressed with the non-linear $H(\mathbf{x})$ function. In ResNet architecture, the part from the input to the output is expressed by the nonlinear function defined as $F(\mathbf{x}) = H(\mathbf{x}) - \mathbf{x}$. The original mapping is repeated as $F(\mathbf{x}) + \mathbf{x}$. That is, the input value $\mathbf{x}$ is arithmetically added to the function $F(\mathbf{x})$. After this process, it is passed through ReLU with the function $F(\mathbf{x}) + \mathbf{x}$ as in Figure 4. In this way, the values in the past layers are strongly transmitted to the next layers [26].

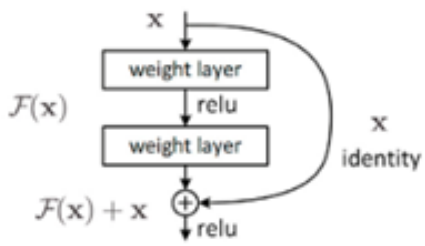


Fig. 4: Residual mapping structure
[26]

3 Model Uncertainty in Bayesian Deep Learning

$\mathcal{D} = \{(\mathbf{x}_1, y_1), \cdots, (\mathbf{x}_N, y_N)\} \subset X \times Y$ is a given dataset where $X \in \mathbb{R}^m$ is a m-dimensional input vector and $Y \in \{1 \cdots C\}$, training instances $\{\mathbf{x}_i, y_i\} \forall X$ are a set of independent and identically distributed (i.i.d.) training samples from some unknown probability measure P on $X \times Y$. The main idea behind $h : X \mapsto Y$ is to get The a hypothesis applying weights of neural net parameters w and a loss function $\ell : Y \times Y \mapsto \mathbb{R}$ as close as possible to the original function that has generated the labels.

Gal et al. [27] introduced a neural network model with prediction time dropout that is similar to a particular variational inference on a Bayesian neural network. The prediction hypothesis uncertainty is approximated by averaging probabilistic feed-forward Monte Carlo dropout sample over the forecast period.

Monte Carlo dropout based inference is a practical method for approximate inference in large and complex models [27]. This prediction is made by training a neural network with dropout before every weight layer. The prediction hypothesis uncertainty is approximated by averaging probabilistic feed-forward Monte Carlo dropout sampling during prediction time. The predictive mean over dropout iterations, T, is used as the final inference,$\hat{y}$, on the test instance $\hat{\mathbf{x}}$.

$$p(\hat{y} = c|\hat{\mathbf{x}}, \mathcal{D}) \approx \hat{\mu}_{pred} = \frac{1}{T} \sum_{y \in T} p(\hat{y}|\theta, \mathcal{D}) \tag{2}$$

where θ is the model weights, $\mathcal{D}$ is the input dataset.

Figure 5 shows the general overview of the Monte Carlo dropout based prediction in the inference time. In the inference time, random neurons in each layer are removing according to the p from the base neural network model to create another model. As a result, T different classification model will be created.

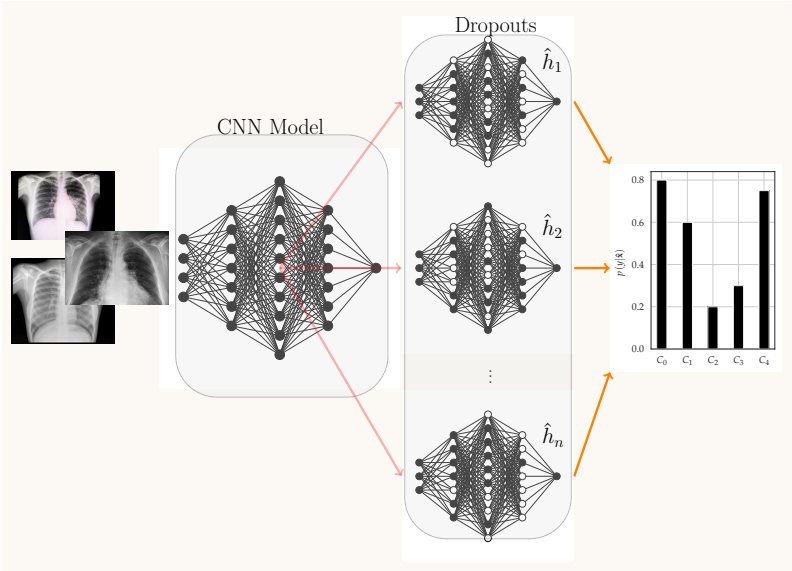


Fig. 5: Illustration of the Monte Carlo dropout based Bayesian prediction.

For each testing input sample $\mathbf{x}$, the predicted label is assigned with the highest predictive mean. We used entropy and predictive variance measures in order to quantify how much the model is uncertain about its labeling predictions. The entropy is a measure defined as the average level of uncertainty fixed in the likely outcomes of a random variable [27].

$$H(\hat{y}|\theta, \mathcal{D}) = -\sum_{c \in C} p\left(f(\mathbf{x}) = c\right) \log p\left(f(\mathbf{x}) = c\right) \tag{3}$$

where $f(\mathbf{x})$ is the classification model prediction, and the p is the probability of the predicted label, which is class c.

$$VAR(\mathbf{y}) = \mathbb{E}\left[\hat{f}(\mathbf{x}|\theta, \mathcal{D}) - \mathbb{E}\left[\hat{f}(\mathbf{x}|\theta, \mathcal{D})\right]\right]^2 \tag{4}$$

4 Data Collection, Explanation

In this work, a publicly available GitHub dataset is used for the model development. The dataset contains chest X-ray and CT images of several lung diseases, including positive, suspected positive of COVID-19 or other viral and bacterial types of diseases such as MERS, SARS, and ARDS. In our previous work [x29], we had only 50 positive and 50 negative X-ray scan images for the training and 20 positive and 20 negative samples for the testing. In this work, we have increased the number

of positive and negative samples for the training and testing sets to 278 and 66, respectively.

Table 1: Chest X-ray labels.

Label	**Number**
Pneumonia/Viral/COVID-19	196
Pneumonia	38
Pneumonia/Fungal/Pneumocystis	21
todo	12
Pneumonia/Bacterial/Streptococcus	12
No Finding	10
Tuberculosis	9
Pneumonia/Viral/SARS	8
Pneumonia/Bacterial/Klebsiella	7
Pneumonia/Bacterial/Mycoplasma	4
Pneumonia/Bacterial/Nocardia	4
Pneumonia/Bacterial/Legionella	4
Pneumonia/Lipoid	4
Pneumonia/Viral/MERS-CoV	4
Pneumonia/Viral/Varicella	4
Pneumonia/Bacterial	2
Pneumonia/Bacterial/Staphylococcus/MRSA	1
Pneumonia/Bacterial/Chlamydophila	1
Pneumonia/Viral/Influenza	1
Pneumonia/Viral/Influenza/H1N1	1
Pneumonia/Fungal/Aspergillosis	1

5 System Model

We executed the experiments 30 times and tested the models for the following question: How accurate are the softmax output prediction's variance and prediction entropy in COVID-19 X-ray images dataset?

As the dataset is not balanced, and the data is not uniform, the traditional accuracy based classification performance evaluation methods is not suitable to assess the models. However, our experiments will provide four standard measurement metrics in information retrieval for evaluating the models such as overall prediction accuracy [28], average recall [29], average precision [30] and F_1 [31].

Precision is the ratio of the true value of the prediction to the prediction error and true value. In Eq. 5., mathematical calculation of precision is illustrated:

$$Precision = \frac{Correct}{Correct + False} \tag{5}$$

Recall is calculated by the fraction of retrieved relevant samples.

$$Precision = \frac{Correct}{Correct + Missed} \tag{6}$$

Firstly, We calculated the precision and recall values for each class and then the mean value. In Eq. 7 and Eq. 8, average precision and recall is presented respectively .

$$Precision_{avg} = \frac{1}{n_{classes}} \sum_{i=0}^{n_{classes}-1} Prec_i \tag{7}$$

$$Recall_{avg} = \frac{1}{n_{classes}} \sum_{i=0}^{n_{classes}-1} Recall_i \tag{8}$$

The F_1 values represents the harmonic mean of precision and sensitivity values.

$$F_1 = 2 \times \frac{Prec_{avg} \times Recall_{avg}}{Prec_{avg} + Recall_{avg}} \tag{9}$$

6 Simulation Results

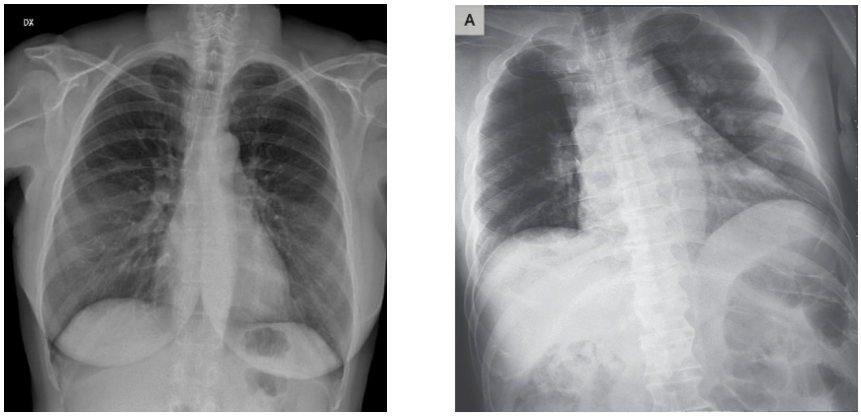
6.1 Dataset

We used the publicly available data set of chest x-rays and CT images of positive or suspected COVID-19 patients for this study. The data set was obtained indirectly from both public sources, physicians and different hospitals. This dataset that contains images and data can be reached and published in the GitHub repository. X-ray images and CT scans are available in the original dataset. We selected X-ray scan images to construct a COVID-19 detection model for our proposed model. While building the model, we used 278 positive and 66 negative COVID-19 posteroanterior X-ray scan images for training and 39 positive and 30 negative images for testing phases.

We rescaled the X-ray scan images to 256×222 size. We have applied different image augmentation methods to create more advanced input instances such as flipping right/left and up/down, rotation and translation utilizing random five various angles.

This dataset is publicly accessible. But, it is important to remark that the number of data sets and the number of samples they include could increase in the future. Besides, we consider the tagging issue. The dataset comprises X-ray scan images containing the disease. There are only three records in the dataset that contain COVID-19 negative samples, and the "finding" field was selected as No finding. Therefore, X-ray images with ARDS and Streptococcus results are labeled to select

COVID-19 negative samples. Additional samples of COVID-19 negative patients will increase in this type of data collection for finding reliable results.



(a) (b)

Fig. 6: An example of two X-ray scan images taken from the dataset (a) with a label of COVID-19 negative, (b) COVID-19 positive.

In order to enhance the performance of the classification model and increase the samples in the data set, we apply image augmentation techniques. The parameters of image augmentation are zoom range 15, rotation range 20, width shift range 0.2, height shift range 0.2, shear range 0.15, horizontal flipping. In Figure 8 an image augmentation technique is presented.

In Fig. 8, it is seen that the first image is the original image and the rest of images are augmented images.

6.2 Experiments

In the experiments, we tested the uncertainty quantification methods and their relations with the classification performance. Simulations were performed on the following machine: 2.8 GHz Quad-Core Intel Core i7 with 16GB of RAM. The experiments were performed using the Python scripts and machine learning libraries: Keras, Tensorflow, and Scikit-learn. Two models, with and without MC dropouts, were trained to obtain COVID-19 detection model's prediction results for the dataset. The hyper-parameters such as the number of hidden layers and the number of neurons in the hidden layers, the activation function, the loss function, and the optimization method are the same for both models. However, the only difference between the two models is the prediction time dropout. In the first model, the drop out is applied at the prediction time, and the second model uses the drop out at the training time.

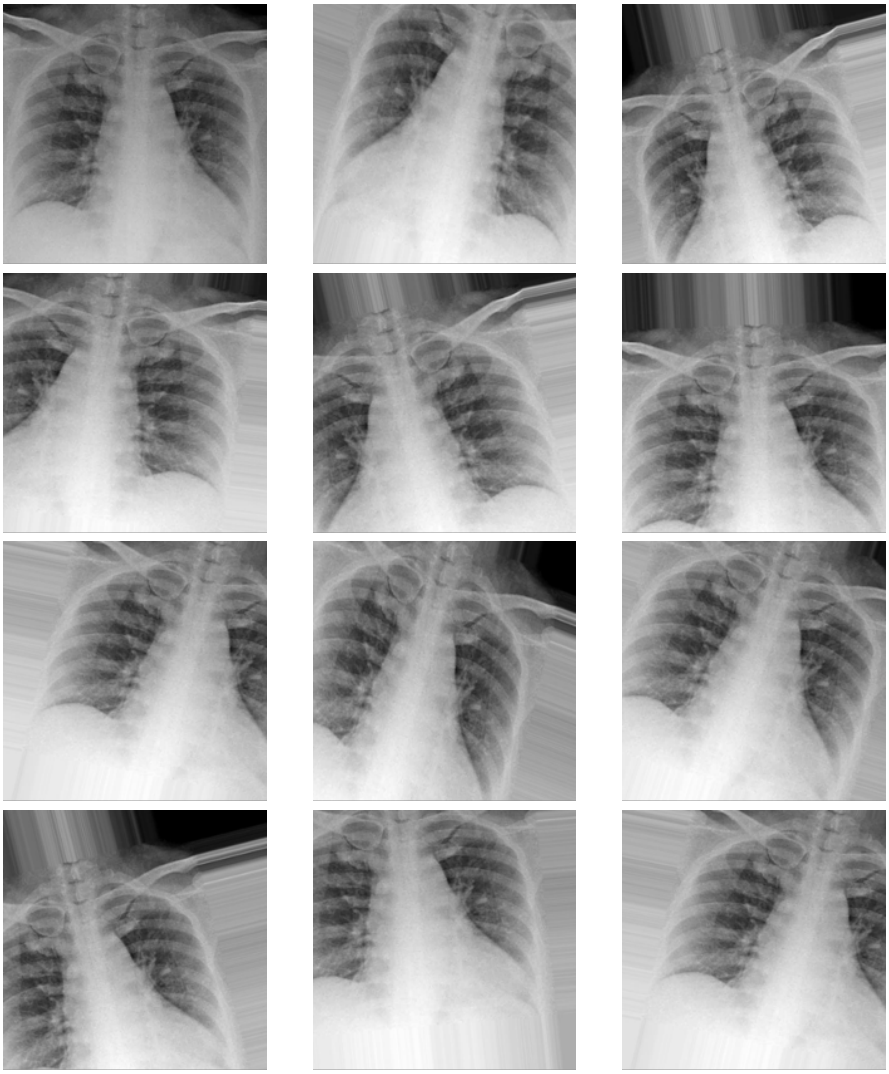


Fig. 7: Example of Data augmentation.

The dataset has been divided into two parts: 33% test sets and 67% training sets. In the test sets, we randomly choose from a subset of the data sets and use them to evaluate the prediction performance of the models. The number of epochs was set to 500, and the epoch size was fixed for all models. Prediction time dropout values were set to 0.0 and increased to 0.9 with an increment of 0.02. Monte Carlo dropout models were used for each test input with an $T = 100$ times for prediction of the test dataset. The predicted labels for the test set are calculated by averaging the output of the last layer's softmax function over T iterations. The prediction uncertainty for

each input in the test sets is quantified using Shannon entropy and prediction variance, respectively. Table 2 shows the classification performance of the Monte Carlo dropout based model and the standard model for each dataset.

Firstly, we used the entropy and prediction variance values of the COVID-19 detection model's softmax probability outputs to compare each uncertainty quantification method to determine the most and least uncertain predictions on the dataset. Figure 8 shows the six most uncertain test instances with entropy-based and softmax prediction probability variance-based uncertainty quantification for the dataset. Each subfigure in the figures shows an input example with prediction probabilities (T times) histogram. According to the figure that shows the highly uncertain examples, softmax probability output distributions for each class show very similar patterns in the results. According to Figure 8, the model's prediction is COVID-19 Positive, and the true label is also COVID-19 Positive for the input example. On the other hand, the uncertainty is very high for the prediction results, which has an uncertainty value of 0.98. Also, the class assignment probabilities for this input example are almost identical for each class label 49.81%, and 50.19%, respectively, for all classes, and it can be difficult reliable. Another example is in Figure 8. The model's class prediction is wrong for this example: it shows the highest probability for all COVID-19 Negative 50.04 %, and the value for the COVID-19 Positive is 49.96% with an uncertainty value of COVID-19 Positive. The class softmax probabilities histogram distribution shows a very similar pattern for each class label.

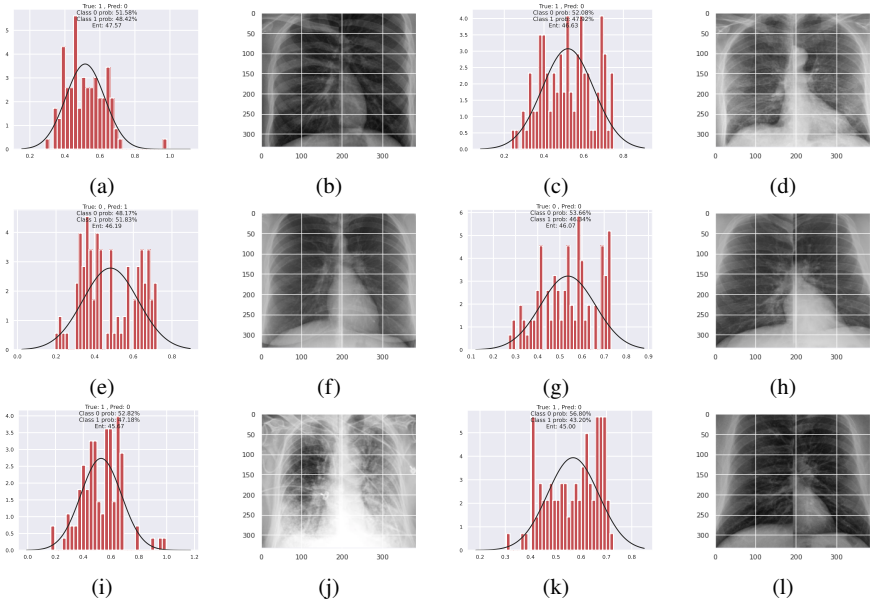


Fig. 8: Most uncertain test instances for the **COVID-19** dataset.

Model	Precision	Recall	F_1	Acc.
VGG-16	0.76	0.75	0.75	0.75
VGG-19	0.74	0.74	0.74	0.74
Resnet-50	0.32	0.57	0.41	0.57
InceptionV3	0.75	0.72	0.71	0.72

Table 2: Performance metrics for CNN pre-trained models for the diagnosis of COVID-19 of the proposed model.

Table 2 provides the pre-trained CNN algorithms used to build detection models for COVID-19. The findings from Table 2 show that pre-trained VGG16 algorithm presents the highest accuracy performance with 75% among the other well-known trained algorithms. The VGG19, ResNet50 and Inception V3 indicate 74%, 57% and 72% respectively. It can also be observed that VGG-16 model achieved the highest F-score value with 75%.

Figure 9 shows the COVID-19 model's entropy-based uncertainty with the $p = 0.2$ value. In the figure, the x-axis shows the model uncertainty value and the y-axis shows the density of data points. The green lines are the true-positive and true-negative values of the classification model. For instance, the sub-figure on the left shows the uncertainty of positive examples (true *COVID-19* labels) predictions with true-positive and false-positive labels, respectively. The true-positive and true-negative labelled instances' uncertainty histogram values are clustered in the lower values (the left side of the figures), which means the low uncertainty in the correctly labelled samples. On the other hand, the false-positive and false-negative labelled instances' uncertainty histogram values are more scattered in the upper values (the right side of the figures), indicating the high uncertainty in the incorrectly labelled samples.

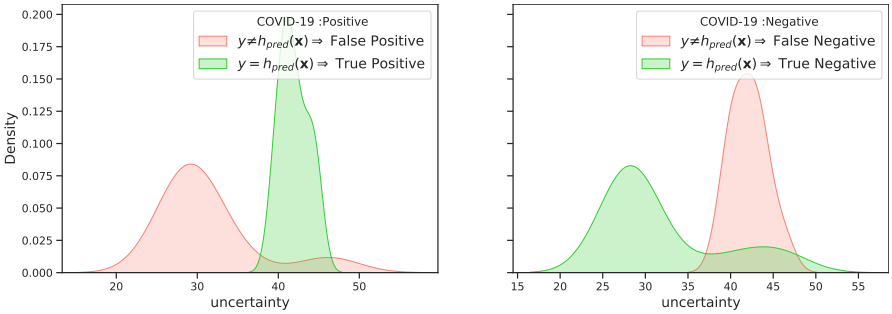


Fig. 9: Bayesian model uncertainty for the COVID-19 dataset.

6.3 Discussion

Our results show that the COVID-19 model provides a precise understanding of the classification model. It shows how the COVID-19 model can be described in an even more comprehensive and well-defined way and the use of the classifier. It uses an extended classification framework to help it describe the classification model, as well as an explanation of the model's structure, more concisely. Figure 9 shows the uncertainty distribution for correct and wrong predictions using the uncertainty framework. According to the figure, the uncertainty expectations are low for the correctly labelled samples, and the incorrect labels are clustered in the upper values (the left side of the figures). This suggests that the model has a more consistent classification (high confidence) for the positive labels (COVID-19 positive); for the incorrectly predicted labels, the model's predictions have low confidence. It concludes that, if a deep learning-based COVID-19 detection system is applicable for the real-case scenario, it is essential to examine the possible prediction uncertainty to catch the wrong predictions.

7 Conclusion

COVID-19 is spreading rapidly around the world. Results of sputum or blood samples tests to detect the presence of this virus in humans are usually available within a few hours or days. For this reason, the importance of faster diagnosis of COVID-19 detection is increasing considerably. In this chapter, we propose uncertainty quantification enhanced transfer learning-based convolutional neural network models by exploiting the VGG-16 model for predicting the presence of COVID- 19 in X-Ray scan images to provide a fully automatic and faster diagnosis. As shown by findings from results, the proposed VGG-16 model detect whether a chest X-ray is COVID-19 or healthy with the highest classification performance as 0.75 accuracy values. The findings suggest that additional data is needed to find proper accuracy performance. Using only a deep learning model in corona detection can lead to irreversible results. The quality of the prediction made by the deep learning model must also be measured. While the studies conducted so far are concerned only with the quality of the model, in this study, the quality of each estimate is also measured.

References

1. L. Wang and A. Wong, "Covid-net: A tailored deep convolutional neural network design for detection of covid-19 cases from chest x-ray images," *arXiv preprint arXiv:2003.09871*, 2020.
2. F. Ucar and D. Korkmaz, "Covidiagnosis-net: Deep bayes-squeezenet based diagnostic of the coronavirus disease 2019 (covid-19) from x-ray images," *Medical Hypotheses*, p. 109761, 2020.

3. I. D. Apostolopoulos and T. A. Mpesiana, “Covid-19: automatic detection from x-ray images utilizing transfer learning with convolutional neural networks,” *Physical and Engineering Sciences in Medicine*, p. 1, 2020.
4. E. E.-D. Hemdan, M. A. Shouman, and M. E. Karar, “Covidx-net: A framework of deep learning classifiers to diagnose covid-19 in x-ray images,” *arXiv preprint arXiv:2003.11055*, 2020.
5. A. Narin, C. Kaya, and Z. Pamuk, “Automatic detection of coronavirus disease (covid-19) using x-ray images and deep convolutional neural networks,” *arXiv preprint arXiv:2003.10849*, 2020.
6. K. Zhang, X. Liu, J. Shen, Z. Li, Y. Sang, X. Wu, Y. Zha, W. Liang, C. Wang, K. Wang, *et al.*, “Clinically applicable ai system for accurate diagnosis, quantitative measurements, and prognosis of covid-19 pneumonia using computed tomography,” *Cell*, 2020.
7. X. Xu, X. Jiang, C. Ma, P. Du, X. Li, S. Lv, L. Yu, Y. Chen, J. Su, and G. Lang, “others," deep learning system to screen coronavirus disease 2019 pneumonia,",” *arXiv preprint arXiv:2002.09334*, 2020.
8. K. Sahinbas and F. O. Catak, “Transfer learning based convolutional neural network for covid-19 detection with x-ray images,” in *Data Science for COVID-19, Computational Perspectives* (U. Kose, D. Gupta, V. H. C. de Albuquerque, and A. Khanna, eds.), ch. 24, pp. 1–24, Elsevier: Elsevier, 2020.
9. P. Michalski, B. Ruszczak, and M. Tomaszewski, “Convolutional neural networks implementations for computer vision,” in *International Scientific Conference BCI 2018 Opole*, pp. 98–110, Springer, 2018.
10. G. Litjens, T. Kooi, B. E. Bejnordi, A. A. A. Setio, F. Ciompi, M. Ghafoorian, J. A. Van Der Laak, B. Van Ginneken, and C. I. Sánchez, “A survey on deep learning in medical image analysis,” *Medical image analysis*, vol. 42, pp. 60–88, 2017.
11. Y. LeCun, B. Boser, J. S. Denker, D. Henderson, R. E. Howard, W. Hubbard, and L. D. Jackel, “Backpropagation applied to handwritten zip code recognition,” *Neural computation*, vol. 1, no. 4, pp. 541–551, 1989.
12. F. Sultana, A. Sufian, and P. Dutta, “Advancements in image classification using convolutional neural network,” in *2018 Fourth International Conference on Research in Computational Intelligence and Communication Networks (ICRCICN)*, pp. 122–129, IEEE, 2018.
13. T. Beysolow II, *Introduction to Deep Learning Using R: A Step-by-Step Guide to Learning and Implementing Deep Learning Models Using R*. Apress, 2017.
14. I. Goodfellow, N. Papernot, P. McDaniel, R. Feinman, F. Faghri, A. Matyasko, K. Hambardzumyan, Y.-L. Juang, A. Kurakin, R. Sheatsley, *et al.*, “cleverhans v0. 1: an adversarial machine learning library,” *arXiv preprint arXiv:1610.00768*, vol. 1, 2016.
15. J. Yosinski, J. Clune, Y. Bengio, and H. Lipson, “How transferable are features in deep neural networks?,” in *Advances in neural information processing systems*, pp. 3320–3328, 2014.
16. L. Zou, S. Yu, T. Meng, Z. Zhang, X. Liang, and Y. Xie, “A technical review of convolutional neural network-based mammographic breast cancer diagnosis,” *Computational and mathematical methods in medicine*, vol. 2019, 2019.
17. S. J. Pan and Q. Yang, “A survey on transfer learning,” *IEEE Transactions on knowledge and data engineering*, vol. 22, no. 10, pp. 1345–1359, 2009.
18. K. Simonyan and A. Zisserman, “Very deep convolutional networks for large-scale image recognition,” *arXiv preprint arXiv:1409.1556*, 2014.
19. X. Zhen, J. Chen, Z. Zhong, B. Hrycushko, L. Zhou, S. Jiang, K. Albuquerque, and X. Gu, “Deep convolutional neural network with transfer learning for rectum toxicity prediction in cervical cancer radiotherapy: a feasibility study,” *Physics in Medicine & Biology*, vol. 62, no. 21, p. 8246, 2017.
20. M. Toğaçar, B. Ergen, and Z. Cömert, “Application of breast cancer diagnosis based on a combination of convolutional neural networks, ridge regression and linear discriminant analysis using invasive breast cancer images processed with autoencoders,” *Medical hypotheses*, vol. 135, p. 109503, 2020.
21. Y. Fu and C. Aldrich, “Using convolutional neural networks to develop state-of-the-art flotation froth image sensors,” *IFAC-PapersOnLine*, vol. 51, no. 21, pp. 152–157, 2018.

22. C. Szegedy, W. Liu, Y. Jia, P. Sermanet, S. Reed, D. Anguelov, D. Erhan, V. Vanhoucke, and A. Rabinovich, "Going deeper with convolutions," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 1–9, 2015.
23. M. Mahdianpari, B. Salehi, M. Rezaee, F. Mohammadimanesh, and Y. Zhang, "Very deep convolutional neural networks for complex land cover mapping using multispectral remote sensing imagery," *Remote Sensing*, vol. 10, no. 7, p. 1119, 2018.
24. F. Chollet, "Building powerful image classification models using very little data," *Keras Blog*, 2016.
25. G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 4700–4708, 2017.
26. K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 770–778, 2016.
27. Y. Gal and Z. Ghahramani, "Dropout as a bayesian approximation: Representing model uncertainty in deep learning," in *international conference on machine learning*, pp. 1050–1059, 2016.
28. I. H. Witten and E. Frank, "Data mining: practical machine learning tools and techniques with java implementations," *Acm Sigmod Record*, vol. 31, no. 1, pp. 76–77, 2002.
29. J. Han, J. Pei, and M. Kamber, *Data mining: concepts and techniques*. Elsevier, 2011.
30. A. Turpin and F. Scholer, "User performance versus precision measures for simple search tasks," in *Proceedings of the 29th annual international ACM SIGIR conference on Research and development in information retrieval*, pp. 11–18, 2006.
31. J. Makhoul, F. Kubala, R. Schwartz, R. Weischedel, *et al.*, "Performance measures for information extraction," in *Proceedings of DARPA broadcast news workshop*, pp. 249–252, Herndon, VA, 1999.

View publication stats